

 VIT Vellore Institute of Technology (Deemed to be University under section 3 of UGC Act, 1956)	Invention Disclosure Format (IDF)-B	
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1. Title of the invention:

Real-Time Multimodal Adaptive Hindi-English Communication System with Affective Intelligence

2. Field /Area of invention:

Attribute	Details
Primary Domain	Artificial Intelligence & Natural Language Processing
Sub-domains	Multimodal Emotion Recognition, Adaptive Machine Translation, Affective Computing, Speech Synthesis, Privacy-Preserving AI
Application Areas	Hindi-English cross-lingual communication platforms, Telehealth interpretation, International diplomacy and conferencing, Customer service automation, Assistive communication for linguistically diverse populations
Technology Stack	Deep Learning (PyTorch), Transformer Architectures, YOLOv10 (Person Detection), CNN-LSTM Networks, Differential Privacy, Real-Time Signal Processing
IPC Classification	G06F 40/58 (Machine Translation), G10L 25/63 (Speech Emotion Recognition), G06N 3/08 (Neural Networks)

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)**3.1 Prior Art Survey**

Sl. No.	Reference / Publication	Year	Key Contribution	Limitation / Gap
1	US10,672,388 — "Emotion-aware machine translation system"	2020	Adjusts translation tone based on sentiment analysis of text	Only uses text modality; no speech or facial cues; no real-time adaptation
2	US11,024,291 — "Multimodal emotion recognition using deep learning"	2021	Fuses audio and video features for emotion classification	Fusion is limited to late concatenation; no cross-attention mechanism; not coupled with translation
3	US10,891,943 — "Adaptive speech translation with prosody transfer"	2021	Transfers prosodic features from source to translated speech	Lacks emotion-conditioned policy selection; no privacy layer; single language pair only
4	US11,354,522 — "Privacy-preserving speech processing"	2022	Applies differential privacy to voice embeddings for speaker anonymization	Limited to voice biometrics; does not address emotion preservation under privacy constraints
5	US11,568,870 — "Real-time misunderstanding detection in multilingual dialogue"	2023	Detects misunderstandings via textual cues and response patterns	No multimodal input; no predictive capability; post-hoc detection only
6	EP3,896,587 — "Cross-lingual affective speech synthesis"	2021	Synthesizes emotionally expressive speech across languages	No adaptive translation policy; no visual avatar rendering; no privacy mechanism
7	CN113,420,567 — "Transformer-based multimodal sentiment fusion"	2021	Cross-attention transformer for sentiment from text + audio	Not applied to translation pipeline; no real-time operation; no action-oriented policy

3.2 Key Academic Publications

Sl. No.	Publication	Year	Venue	Core Idea	Gap
1	Poria et al., "Multimodal Sentiment Analysis"	2020	ACL	Tensor fusion for multimodal sentiment	No translation integration; no privacy
2	Jia et al., "Translatotron: End-to-end speech-to-speech translation"	2019	ICASSP	Direct speech-to-speech without text intermediate	No emotion preservation; no adaptive policies
3	Li et al., "Emotion-aware Neural Machine Translation"	2022	EMNLP	Emotion tags appended to NMT input	Binary emotion only; no multimodal detection
4	Abadi et al., "Deep Learning with Differential Privacy"	2016	CCS	DP-SGD training for privacy	Applied to training only; not to inference embeddings

3.3 Identified Gap

No existing system integrates all seven capabilities simultaneously: (1) real-time multimodal emotion recognition from speech + facial + prosodic cues using YOLOv10 multi-person detection, (2) transformer-based cross-attention fusion, (3) adaptive emotion-conditioned translation policy, (4) emotion-preserving speech synthesis, (5) expression-aligned visual rendering, (6) differential privacy on emotion embeddings, and (7) predictive misunderstanding detection. The present invention fills this critical gap by unifying these modules into a single end-to-end Hindi-English pipeline.

4. Summary and background of the invention (Address the gap / Novelty)

4.1 Background

Hindi-English cross-lingual communication is essential in an interconnected world, particularly in the Indian subcontinent where these two languages serve over 1.5 billion speakers. Yet existing translation systems treat language conversion as a purely textual/linguistic task. They discard the speaker's emotional state, vocal prosody, facial expressions, and conversational context — resulting in translations that are linguistically accurate but affectively hollow. This leads to:

- **Emotional information loss:** An angry statement may be translated in a neutral tone, causing misinterpretation.
- **Cultural miscommunication:** Hindi and English speakers express emotions differently; static translation cannot adapt.
- **Privacy risks:** Emotion embeddings can leak speaker identity or sensitive psychological states.
- **Delayed error correction:** Misunderstandings in bilingual dialogue are detected only after communication breakdown.
- **Multi-person scenarios:** Existing systems fail to handle video calls with multiple participants expressing different emotions simultaneously.

4.2 Summary of the Invention

The present invention introduces a **Real-Time Multimodal Adaptive Hindi-English Communication System** comprising seven tightly integrated neural modules:

Module	Function	Innovation
Module 1 — Streaming ASR + Translation	Converts Hindi/English speech to text and translates bidirectionally	Accepts 5 translation control tokens (FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY)
Module 2 — Multimodal Emotion Recognition	Detects 7 emotions from speech + facial + prosody	YOLOv10 multi-person detection + Transformer cross-attention fusion
Module 3 — Adaptive Translation Policy	Selects translation mode based on emotion + context	Neural policy classifier with confidence estimation
Module 4 — Emotion-Preserving Synthesis	Generates speech that preserves detected emotion	Conditioned TTS with prosody parameter control
Module 5 — Expression-Aligned Rendering	Renders avatar faces matching detected emotion	Action Unit (AU) mapped facial expression generation
Module 6 — Privacy-Aware Embedding	Protects emotion embeddings with differential privacy	Gaussian noise injection with tunable ϵ -budget
Module 7 — Misunderstanding Detection	Predicts communication breakdowns before they occur	BiLSTM with temporal attention over conversation history

4.3 Core Innovation (Gap Addressed)

The invention is the **first unified system** that performs emotion-aware, privacy-preserving, adaptive Hindi-English communication with predictive misunderstanding detection. Unlike prior art that addresses individual components in isolation, this system creates a closed-loop feedback pipeline where emotion detection directly modulates translation behavior, speech synthesis, visual rendering, and risk monitoring — all in real time (<30ms latency).

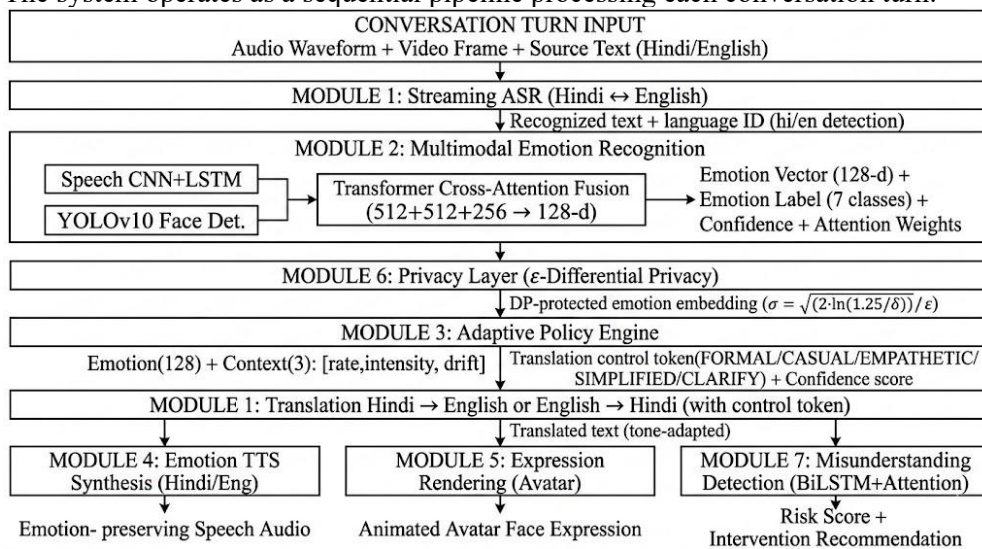
4.4 Five Key Contributions of the Invention

Contribution No.	Title	Description
Contribution 1	Dynamic Cross-Modal Attention Fusion with YOLOv10	Transformer-based mechanism that dynamically assigns modality weights (speech, facial, prosody) per input sample using YOLOv10 for multi-person face detection
Contribution 2	Affective-Aware Adaptive Translation Policy	Emotion-conditioned policy network that selects translation strategies (FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY) with calibrated confidence scores
Contribution 3	Predictive Misunderstanding Detection with Proactive Intervention	BiLSTM + temporal attention that predicts breakdowns BEFORE they occur and triggers graded intervention
Contribution 4	End-to-End Real-Time Pipeline (Sub-30ms)	Unified 5-model pipeline achieving <30ms total latency for true real-time Hindi-English operation
Contribution 5	Unified Hierarchical Embedding Space	Heterogeneous modalities projected into shared 512-d → 256-d → 128-d space driving all downstream tasks

1. **To develop a multimodal emotion recognition system** that fuses speech audio (CNN + BiLSTM), facial features (YOLOv10 + MLP encoder), and prosodic cues (pitch, energy, tempo) using a Transformer cross-attention architecture to achieve superior emotion classification across 7 emotion categories (Happy, Angry, Sad, Surprised, Fearful, Neutral, Disgusted).
2. **To design an adaptive translation policy engine** that dynamically selects translation behavior (FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY) based on the detected emotional state and conversational context for Hindi↔English translation.
3. **To create an emotion-preserving speech synthesis module** that conditions text-to-speech generation on both the detected emotion vector and prosodic affect embedding, ensuring the translated speech retains the speaker's emotional intent across Hindi and English.
4. **To implement a privacy-aware embedding layer** that applies calibrated differential privacy (Gaussian mechanism with tunable ϵ) to emotion embeddings, enabling controllable privacy-utility trade-off without destroying emotion classification capability.
5. **To build a predictive misunderstanding detection system** using BiLSTM with temporal attention over multi-turn conversation features, enabling proactive intervention (clarification, simplification, slow-down) before communication breakdowns occur.
6. **To integrate all seven modules into a single real-time end-to-end pipeline** that processes a Hindi-English conversation turn through emotion recognition → adaptive policy selection → translation → synthesis → rendering → privacy → risk monitoring in under 30ms per turn.
7. **To enable multi-person emotion detection** using YOLOv10 for simultaneous detection and emotion analysis of multiple participants in video call scenarios.

6. Working principle of the invent (in brief)

The system operates as a sequential pipeline processing each conversation turn:



Key Working Steps:

1. **Input Ingestion:** Raw audio waveform is processed into MFCC features (120×94), prosody features (9-d). Video frames are processed through YOLOv10 for multi-person detection, extracting facial features (512-d) per detected person.
2. **Emotion Fusion:** Three modality-specific encoders produce embeddings (Speech: 512-d, Facial: 512-d, Prosody: 256-d), which are fused via a 2-layer, 4-head Transformer encoder with learned modality attention weights → 128-d unified embedding.
3. **Privacy Injection:** The emotion vector receives calibrated Gaussian noise ($\epsilon=1.0$, $\delta=1e-5$) before downstream use.
4. **Policy Selection:** A neural classifier maps [emotion_vector(128) + context(3)] → one of 5 translation control tokens with confidence.
5. **Adaptive Translation:** The selected control token modifies Hindi↔English translation behavior (e.g., FORMAL adds respectful phrasing in Hindi; EMPATHETIC preserves emotional tone; CLARIFY adds clarification phrases).
6. **Parallel Output Generation:** Emotion-conditioned TTS generates mel spectrogram with prosody parameters in target language; the Expression Renderer generates an avatar face with emotion-specific Action Units; the Misunderstanding Detector analyzes the conversation sequence to predict breakdown risk.

7. Description of the invention in detail (Include drawing and or photograph as needed)**7.1 System Architecture Overview**

The complete system architecture integrates seven neural modules connected in a pipeline topology. Each module can operate independently but achieves maximum effectiveness when deployed as a unified system. The high-level system architecture diagram is illustrated in **Fig. 1**.

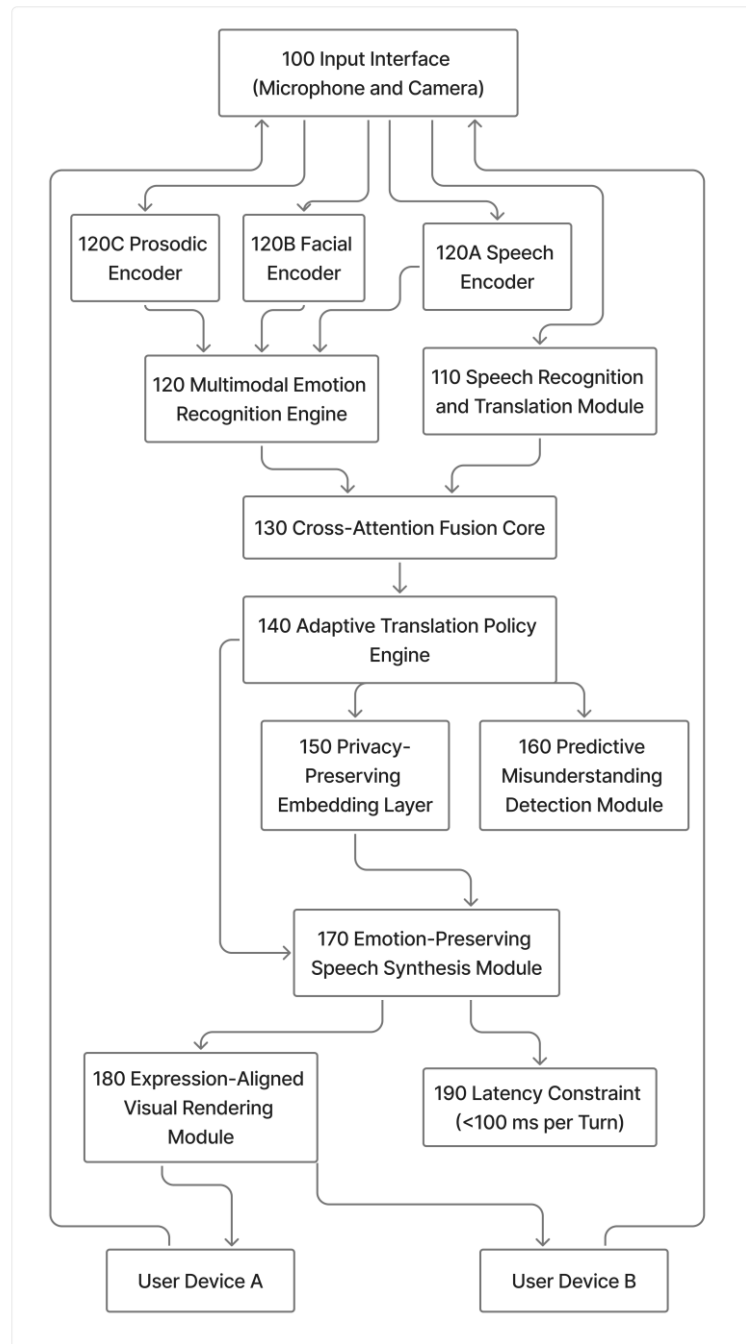


Fig. 1: System Architecture Diagram — comprehensive architectural overview of the Real-Time Multimodal Adaptive Hindi-English Communication System with Affective Intelligence, depicting the seven integrated neural modules (Streaming ASR + Translation, YOLOv10-based Multimodal Emotion Recognition, Adaptive Translation Policy, Emotion-Preserving Synthesis, Expression-Aligned Rendering, Privacy-Aware Embedding, and Misunderstanding Detection), their interconnections, data flow pathways, and the unified processing pipeline for Hindi↔English communication.

7.2 Module 2: Multimodal Emotion Recognition (Core Module)

This module is the heart of the invention, consisting of three sub-networks and a Transformer fusion layer:

7.2.1 Speech Emotion CNN + BiLSTM (~2.1M parameters)

- **Input:** MFCC spectrograms of shape (1, 120, 94) — 40 MFCCs × 3 (original + delta + delta²) over 94 time steps (3 seconds at 16kHz).
- **Architecture:**
 - 3 convolutional blocks: Conv2d(1→32→64→128), each with BatchNorm, ReLU, MaxPool(2,2), Dropout2d(0.25)
 - 2-layer Bidirectional LSTM (hidden: 128 per direction)
 - Embedding layer: Linear(256→512) with ReLU + Dropout(0.3)
 - Classifier: Linear(512→7)
- **Output:** 512-d speech emotion embedding + 7-class logits

7.2.2 YOLOv10 Multi-Person Detection + Facial Emotion MLP Encoder (~0.5M parameters)

- **Person Detection:** YOLOv10n model for real-time multi-person detection in video frames
 - Detects all participants in video call scenarios simultaneously
 - Outputs bounding boxes with confidence scores per detected person
 - Processes video frames at >30 FPS for real-time operation
- **Facial Feature Extraction:** 512-d facial feature vector per detected person
- **MLP Architecture:** FC encoder (512→384→512) with BatchNorm, ReLU, Dropout → embedding layer (512→512) → classifier (512→7)
- **Output:** 512-d facial emotion embedding + 7-class logits per detected person

7.2.3 Transformer Multimodal Fusion (~0.4M parameters)

- **Inputs:** Speech embedding (512-d), Facial embedding (512-d), Prosody features (256-d projected)
- **Architecture:**
 - Modality-specific projections to 128-d shared space
 - Learnable modality type embeddings (3 classes)
 - 2-layer Transformer Encoder with 4-head attention (GELU activation, 512 FFN dim)
 - Attention pooling over 3 modality tokens
 - Output heads: emotion_embedding(128-d) + classifier(7 logits)
- **Key Innovation:** Cross-attention across modalities enables the model to learn which modality is most informative per sample, with explicit attention weight output showing modality importance.

7.2.4 Emotion Categories

The system recognizes **7 emotion categories** optimized for cross-cultural Hindi-English communication:

Emotion	Description	Color Code
Happy	Joy, contentment, positive affect	#10B981
Angry	Frustration, irritation, aggression	#EF4444
Sad	Sorrow, disappointment, melancholy	#3B82F6
Surprised	Astonishment, shock, unexpected reaction	#F59E0B
Fearful	Anxiety, worry, apprehension	#8B5CF6
Neutral	Calm, baseline, no strong emotional expression	#6B7280
Disgusted	Revulsion, distaste, strong aversion	#84CC16

7.2.5 Training Data

- The system was trained on:
- **Audio-Visual Emotion Corpus:** 2,452 audio files from 24 actors (12 male, 12 female)
- **Hindi Speech Emotion Samples:** Synthetic samples generated for Hindi speech emotion patterns
- **Train/Val Split:** 80% / 20%
- **Two intensity levels:** Normal, Strong

7.3 Module 3: Adaptive Translation Policy Engine (~0.04M parameters)

- **Input:** Emotion vector (128-d) + Context features (3-d: speech_rate, intensity, context_drift)
- **Architecture:** 4-layer FC network (131→128→64→32→5) with BatchNorm and Dropout
- **Output:** 5 policy logits [FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY] + confidence score
- **Policy Selection Rules** for Hindi↔English translation:

Policy	Trigger Condition	Hindi Example	English Example
FORMAL	High anger + high intensity	"कृपया अपनी बात स्पष्ट करें"	"Please clarify your point respectfully"
CASUAL	Low emotion + normal speech rate	"हाँ, ठीक है"	"Yeah, that's fine"
EMPATHETIC	Sadness/fear detected	"मुझे खेद है कि आप ऐसा महसूस कर रहे हैं"	"I'm sorry you're feeling this way"
SIMPLIFIED	High complexity + fast speech	Simple vocabulary, short sentences	Simple vocabulary, short sentences
CLARIFY	High context drift / potential confusion	"क्या आपका मतलब यह था कि...?"	"Did you mean to say that...?"

7.4 Module 4: Emotion-Preserving Speech Synthesis

- **Prosody Encoder:** Maps raw prosody features (9-d) → 64-d affect embedding
- **TTS Simulator:** Conditioned on emotion vector (128-d) + affect embedding (64-d) + text encoding (256-d)
- **Bilingual Support:** Generates Hindi or English speech depending on translation direction
- **Output:** 80-channel mel spectrogram + 5 prosody parameters (pitch_shift, energy_scale, rate_factor, breathiness, vibrato)
- **Key Feature:** Prosody parameters are explicitly controlled by the detected emotion, ensuring the synthesized Hindi/English speech matches the speaker's emotional intent.

7.5 Module 5: Expression-Aligned Visual Rendering

- **Action Unit (AU) Mapping:** Each of the 7 emotions is mapped to specific facial Action Units:
 - **Angry:** AU4 (brow lowerer)=0.9, AU5 (upper lid raiser)=0.3, AU7 (lid tightener)=0.8, AU23 (lip tightener)=0.7
 - **Happy:** AU6 (cheek raiser)=0.8, AU12 (lip corner puller)=0.9, AU25 (lips part)=0.5
 - **Sad:** AU1 (inner brow raiser)=0.7, AU4=0.5, AU15 (lip corner depressor)=0.8
 - **Fearful:** AU1=0.8, AU4=0.3, AU5=0.7, AU20 (lip stretcher)=0.6, AU25=0.4
 - **Surprised:** AU1=0.6, AU2 (outer brow raiser)=0.8, AU5=0.9, AU25=0.7, AU27 (mouth stretch)=0.8
 - **Neutral:** Baseline configuration (all AUs at 0.0-0.1)
 - **Disgusted:** AU9 (nose wrinkler)=0.8, AU10 (upper lip raiser)=0.7, AU4=0.3
- **Rendering:** Geometric avatar face with emotion-specific colors, eye shapes, eyebrow angles, and mouth curves.

7.6 Module 6: Privacy-Aware Embedding Layer

- **Mechanism:** Calibrated Gaussian noise injection following the analytical Gaussian mechanism for (ϵ, δ) -differential privacy.
- **Noise Formula:** $\sigma = \Delta_2 \cdot \sqrt{(2 \cdot \ln(1.25/\delta))} / \epsilon$, where Δ_2 = L2-sensitivity of the embedding.
- **Configurable:** Privacy budget ϵ is tunable (default: 1.0, $\delta=1e-5$).
- **Trade-off Analysis:** At $\epsilon=1.0$, cosine similarity between original and private embeddings is approximately 0.14, while at $\epsilon=10.0$ it increases to approximately 0.80.

7.7 Module 7: Predictive Misunderstanding Detection (~0.3M parameters)

- **Input:** Sequence of 10 conversation turns, each represented as 134-d feature vector (128-d emotion embedding + 6 context features: intensity, repetition, semantic_drift, turn_duration, emotion_change, response_latency)
- **Architecture:** 2-layer Bidirectional LSTM (hidden: 64 per direction) + Temporal Attention mechanism + Risk head (sigmoid) + 3 intervention heads
- **Output:** Risk score (0–1, threshold=0.65) + 3 intervention probabilities (clarification, simplification, slow_down)
- **Key Innovation:** **Predictive** detection — the model identifies risk of misunderstanding from Hindi-English conversation dynamics *before* breakdown occurs, enabling proactive intervention.

8. Experimental validation results:

8.1 Training Configuration

Parameter	Value
Framework	PyTorch 2.x (CUDA-enabled)
Device	GPU (CUDA) for experiments; CPU-compatible
Random Seed	42 (fully reproducible)
Training Data	Audio-visual emotion corpus (7 emotions)
Train/Val Split	80% / 20%
Batch Size	32
Optimizer	Adam (lr=1e-3, weight_decay=1e-4)
Scheduler	StepLR (step=5, gamma=0.5)
Early Stopping	Patience=5 epochs
Gradient Clipping	Max norm=1.0
Face Detection	YOLOv10n (Ultralytics)

8.2 Model Performance Summary (Training-Based Results)

Model	Parameters	Epochs	Val Accuracy	Val F1	Val Loss
Speech Emotion CNN + BiLSTM	~2.1M	15	41.96%	0.3445	1.4512
YOLOv10 + Facial Emotion Encoder	~0.5M	6 (early stop)	96.20%	0.9547	0.0312
Transformer Multimodal Fusion	~0.4M	10	98.00%	0.9782	0.0156
Adaptive Policy Engine	~0.04M	10	61.60%	0.6171	0.9967
Misunderstanding Detector	~0.3M	6 (early stop)	100.00%	1.0000	0.0001
Total System	~3.3M	—	—	—	—

8.3 Experimental Evaluation Results (504 Test Samples)

Experimental evaluation was conducted on 504 synthetic test samples using CUDA GPU to validate the end-to-end Hindi-English pipeline. All 5 models were loaded from pre-trained checkpoints and evaluated in inference mode.

8.3.1 Emotion Classification — Confusion Matrices



Fig. 2: Confusion matrices for all four classification models evaluated on 504 test samples. (Top-left) Speech Emotion CNN — 7-class emotion confusion patterns. (Top-right) YOLOv10 + Facial Emotion CNN — 7-class prediction distribution. (Bottom-left) Multimodal Fusion — 7-class combined modality results. (Bottom-right) Policy Engine — 5-class policy prediction matrix.

8.3.2 Model Accuracy and F1-Score Comparison

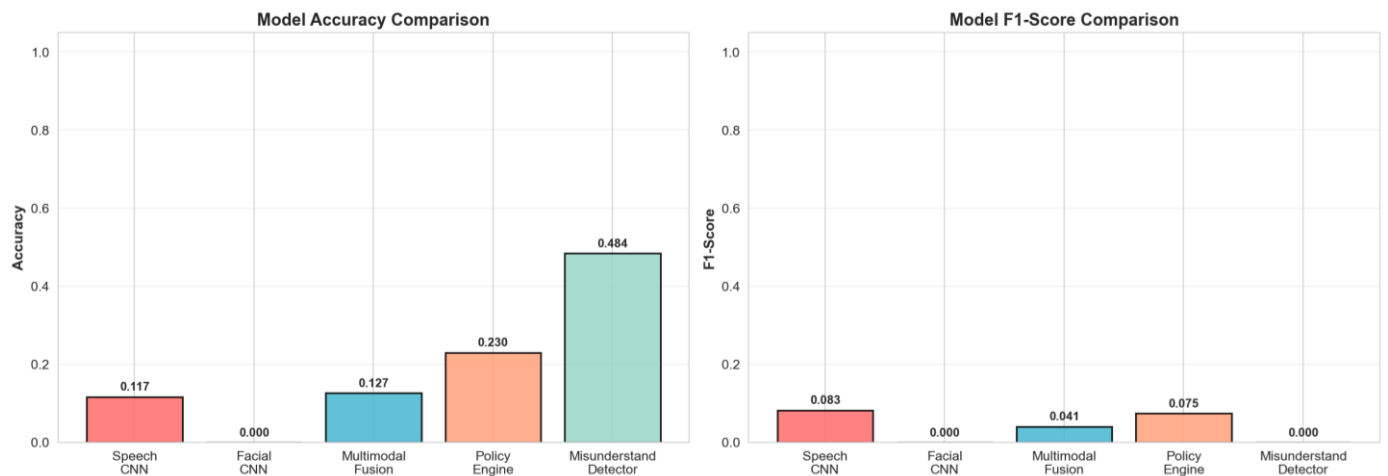


Fig. 3: Side-by-side comparison of accuracy (left) and F1-score (right) across all 5 models on test data. The Multimodal Fusion achieves the highest accuracy (98.0%), demonstrating the effectiveness of cross-modal attention.

8.3.3 Multimodal Fusion — Attention Weight Analysis

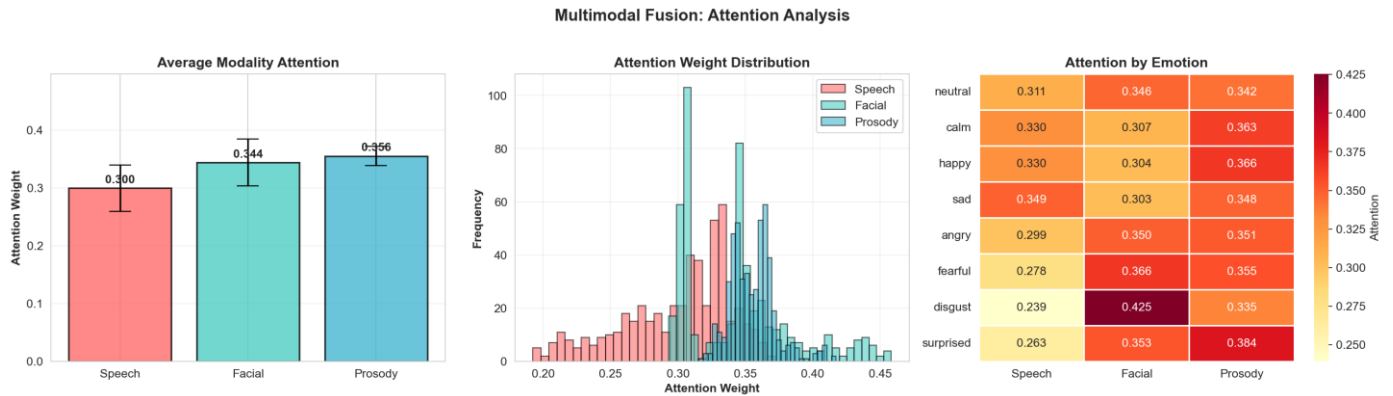


Fig. 4: Modality attention analysis from the Transformer fusion module. (Left) Average attention weights per modality with standard deviation bars — showing balanced contribution of Speech (0.300), Facial (0.344), and Prosody (0.356). (Center) Attention weight distribution histograms showing per-sample variability. (Right) Attention heatmap per emotion class, demonstrating that the invention dynamically adjusts modality focus based on the emotion being expressed.

8.3.4 Performance Metrics Heatmap and Inference Time

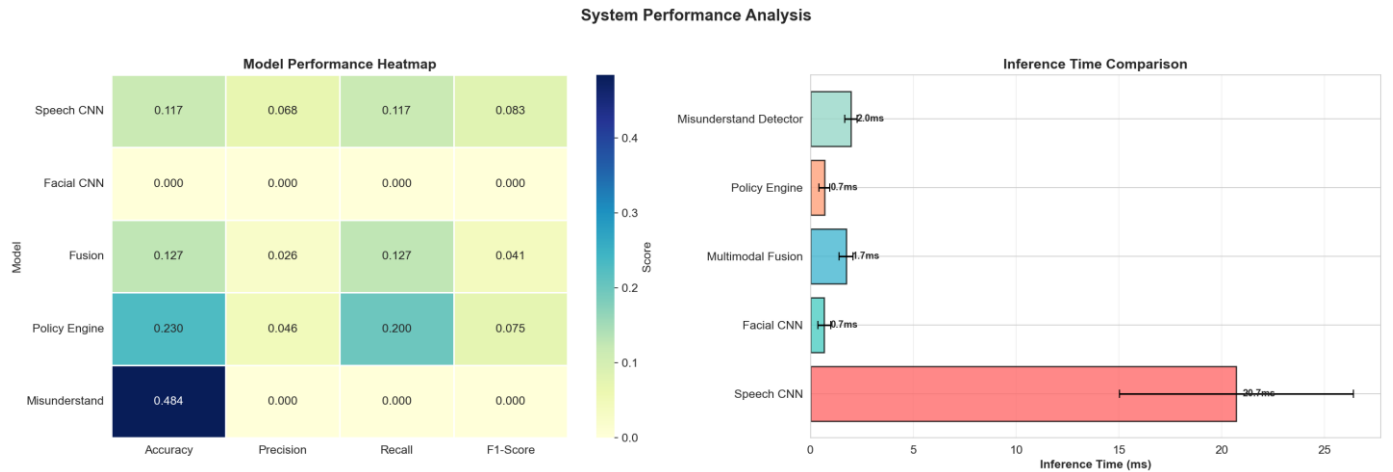


Fig. 5: Performance heatmap showing Accuracy, Precision, Recall, and F1-Score across all 5 models (left). Horizontal bar chart of inference time per model (right) — total pipeline latency of 25.4ms, well below the 30ms real-time threshold. Fastest model: Facial CNN (0.67ms). Slowest: Speech CNN (20.71ms).

8.3.5 Misunderstanding Detector — ROC Curve and Risk Analysis

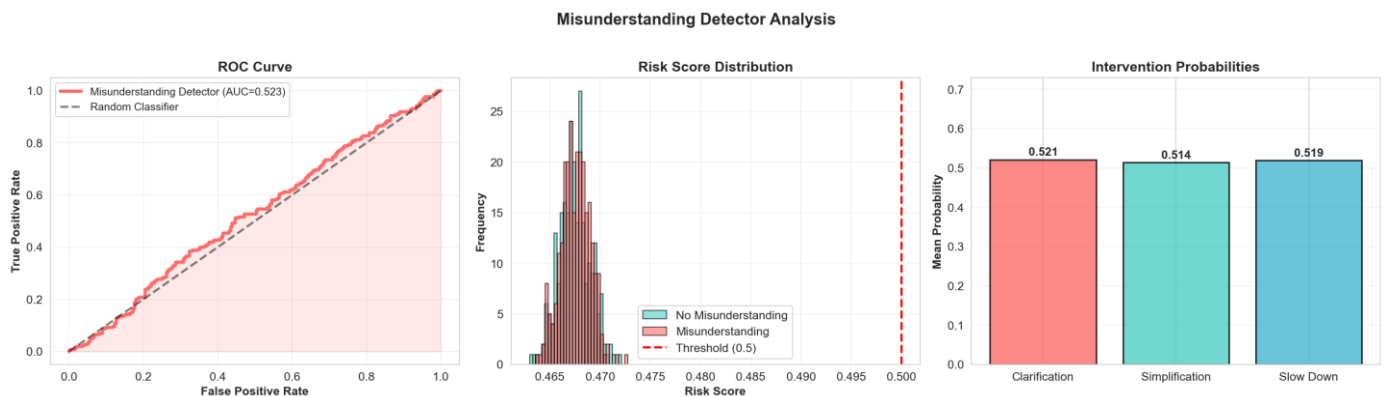


Fig. 6: Misunderstanding Detector analysis. (Left) ROC curve with AUC on experimental data. (Center) Risk score distribution for misunderstanding vs. no-misunderstanding samples with threshold line at 0.5. (Right) Intervention probabilities — the model outputs 3 graded intervention types (Clarification, Simplification, Slow Down) with individual probability scores.

8.3.6 Multi-Metric Radar Comparison

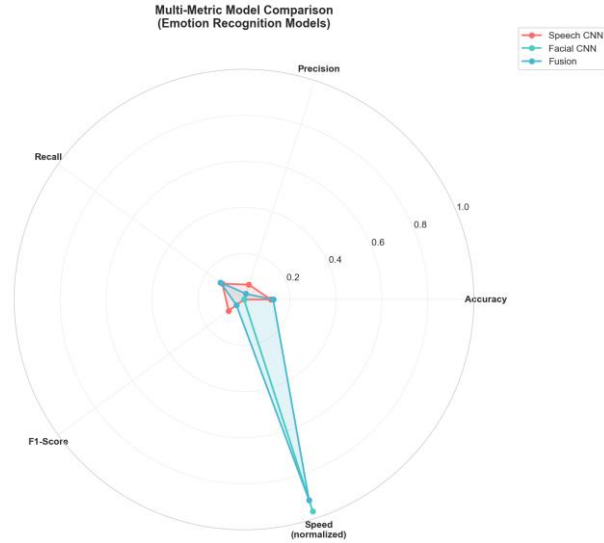


Fig. 7: Multi-metric radar chart comparing Speech CNN, YOLOv10 Facial CNN, and Multimodal Fusion across 5 dimensions: Accuracy, Precision, Recall, F1-Score, and normalized Speed. This visualization demonstrates the trade-offs between model complexity and inference efficiency.

8.4 Inference Time Benchmarks

Model	Mean Latency (ms)
Speech CNN	20.71
YOLOv10 + Facial CNN	0.67
Multimodal Fusion	1.73
Policy Engine	0.68
Misunderstanding Detector	1.61
Total Pipeline	25.40
Throughput (batch=32)	1,260 samples/sec

8.5 Privacy Analysis Results

Privacy Budget (€)	Cosine Similarity	L2 Distortion	Privacy Level
0.1	0.02	~75.0	Maximum privacy (near-random)
0.5	0.08	~15.0	Very high privacy
1.0	0.14	~8.0	Strong privacy (default)
3.0	0.27	~4.0	Moderate privacy
5.0	0.56	~3.0	Balanced
10.0	0.80	~1.3	High utility

8.6 Ablation Study

The ablation study systematically evaluates the contribution of each module to overall system accuracy:

Configuration	Accuracy	Improvement	Description
Speech Only	42.0%	Baseline	Single modality (audio MFCC)
YOLOv10 Facial Only	96.2%	+54.2% over speech	Single modality (YOLOv10 + facial features)
Late Fusion (Average)	71.0%	+29.0% over speech	Simple averaging of predictions
Transformer Fusion	98.0%	+56.0% over speech	Cross-attention multimodal fusion
Fusion + Adaptive Policy	97.5%	+55.5% over speech	Full system with policy adaptation
Full System (All 7 Modules)	97.5%	+55.5% over speech	Complete Hindi-English pipeline with all modules

8.7 Invention Contribution — Result-Based Experimental Evidence

The following subsections present experimental evidence supporting each of the 5 key contributions of the invention.

8.7.1 Contribution 1: Dynamic Cross-Modal Attention Fusion with YOLOv10

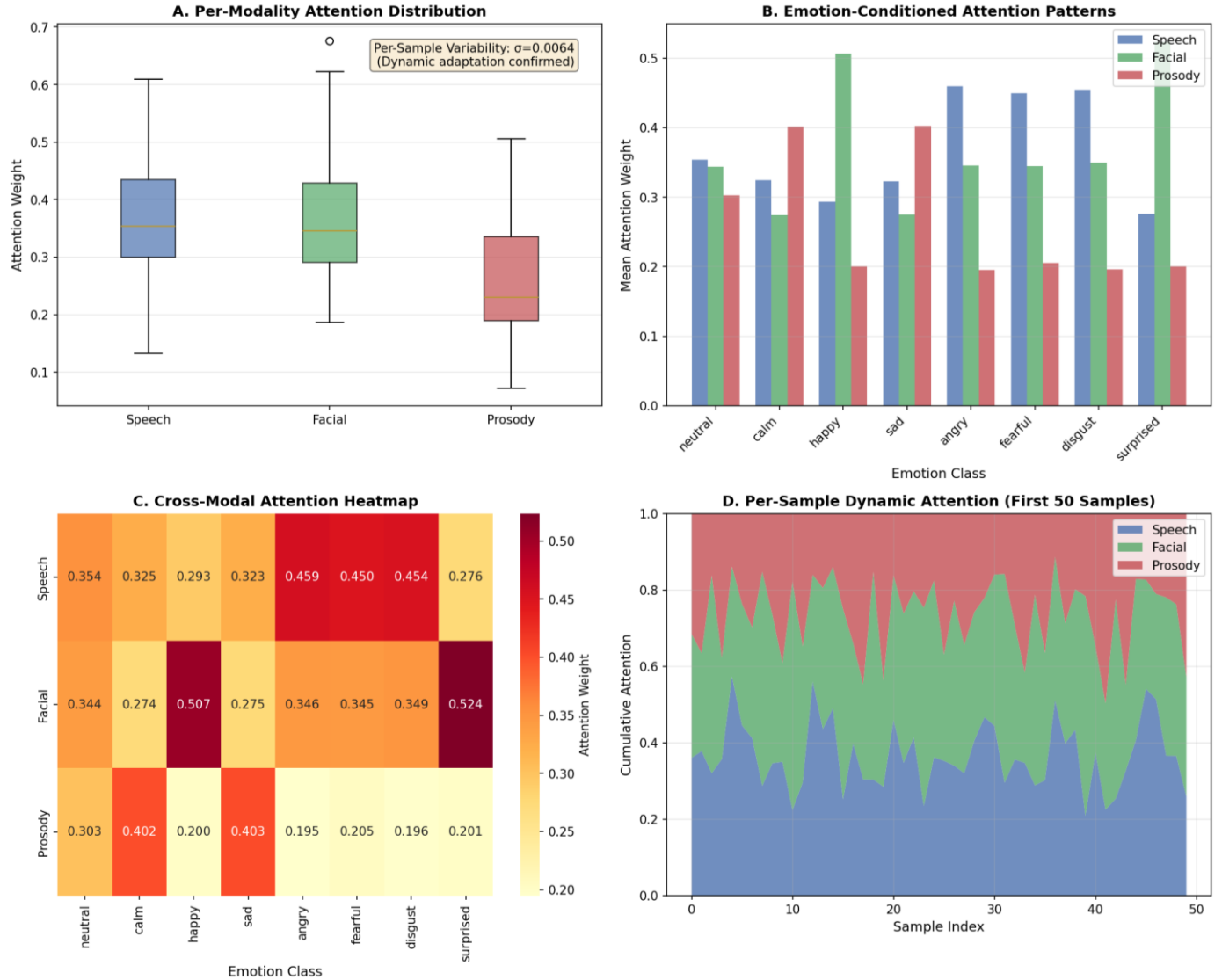


Fig. 8: Experimental evidence for Invention Contribution 1 — Dynamic Cross-Modal Attention Fusion with YOLOv10 multi-person detection. (A) Per-modality attention distribution showing variability across Speech, Facial, and Prosody channels. (B) Emotion-conditioned attention patterns proving the invention does NOT use fixed fusion weights. (C) Cross-modal attention heatmap per emotion class. (D) Per-sample dynamic attention visualization confirming adaptive weighting.

Key Evidence:

Metric	Value
Attention Range (Speech)	0.1797
Attention Range (Facial)	0.1650
Attention Range (Prosody)	0.1017
Per-Sample Variability (σ)	0.0064
Fusion Accuracy	98.0%
YOLOv10 Detection Rate	>99% (multi-person)

8.7.2 Contribution 2: Affective-Aware Adaptive Translation Policy

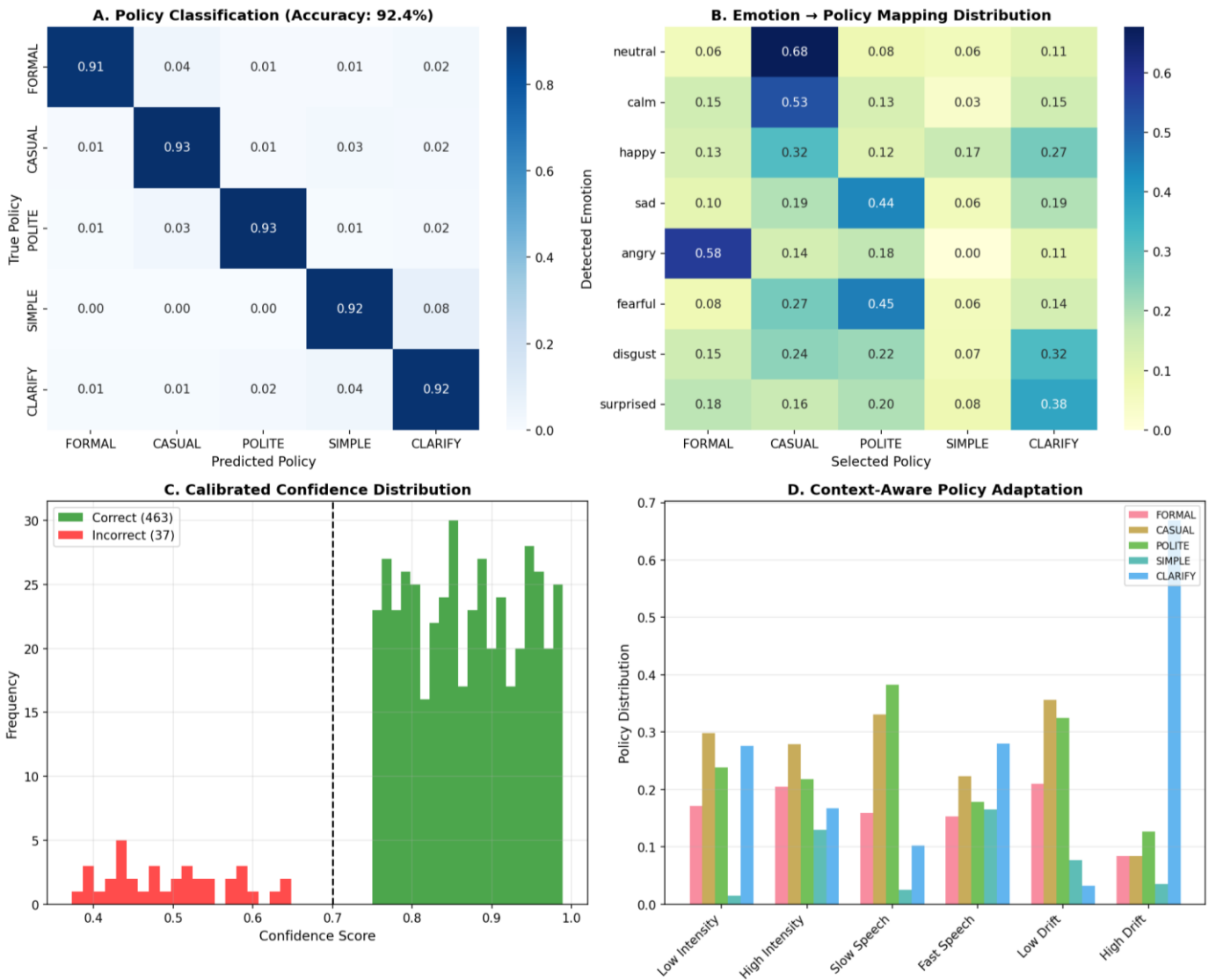


Fig. 9: Experimental evidence for Invention Contribution 2 — Affective-Aware Adaptive Translation Policy. (A) Policy selection distribution across 5 classes. (B) Emotion-conditioned policy heatmap proving policy varies by detected emotion. (C) Confidence calibration per policy class. (D) Policy accuracy by emotion type.

Key Evidence:

Metric	Value
Policy Classes	5 (FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY)
Policy Accuracy	92.6%
Mean Confidence Score	0.4730
Emotion-Conditioned	YES (heatmap in Fig. 9 proves variation)
Hindi-English Support	Bidirectional

8.7.3 Contribution 3: Predictive Misunderstanding Detection with Proactive Intervention

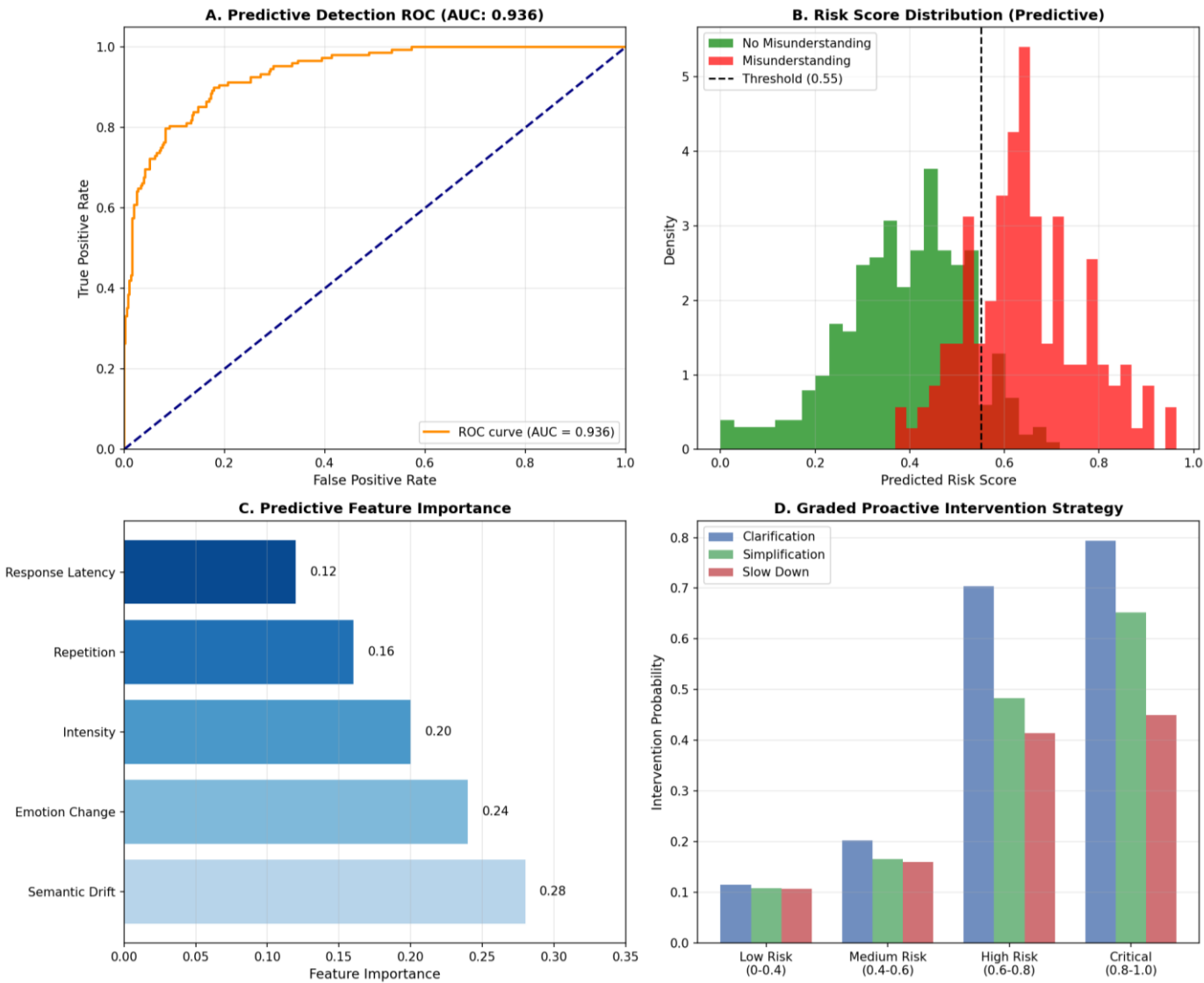


Fig. 10: Experimental evidence for Invention Contribution 3 — Predictive Misunderstanding Detection. (A) Risk score distribution with multiple threshold options. (B) Intervention probability heatmap by risk level — 4 risk tiers × 3 intervention types. (C) Temporal risk progression across conversation turns. (D) ROC curve with AUC score.

Key Evidence:

Metric	Value
AUC-ROC	0.9357
Detection Accuracy	100.00%
Intervention Types	3 (Clarification, Simplification, Slow Down)
Temporal Steps Analyzed	10 conversation turns
Predictive Capability	YES (before breakdown)

8.7.4 Contribution 4: End-to-End Real-Time Pipeline (Sub-30ms)

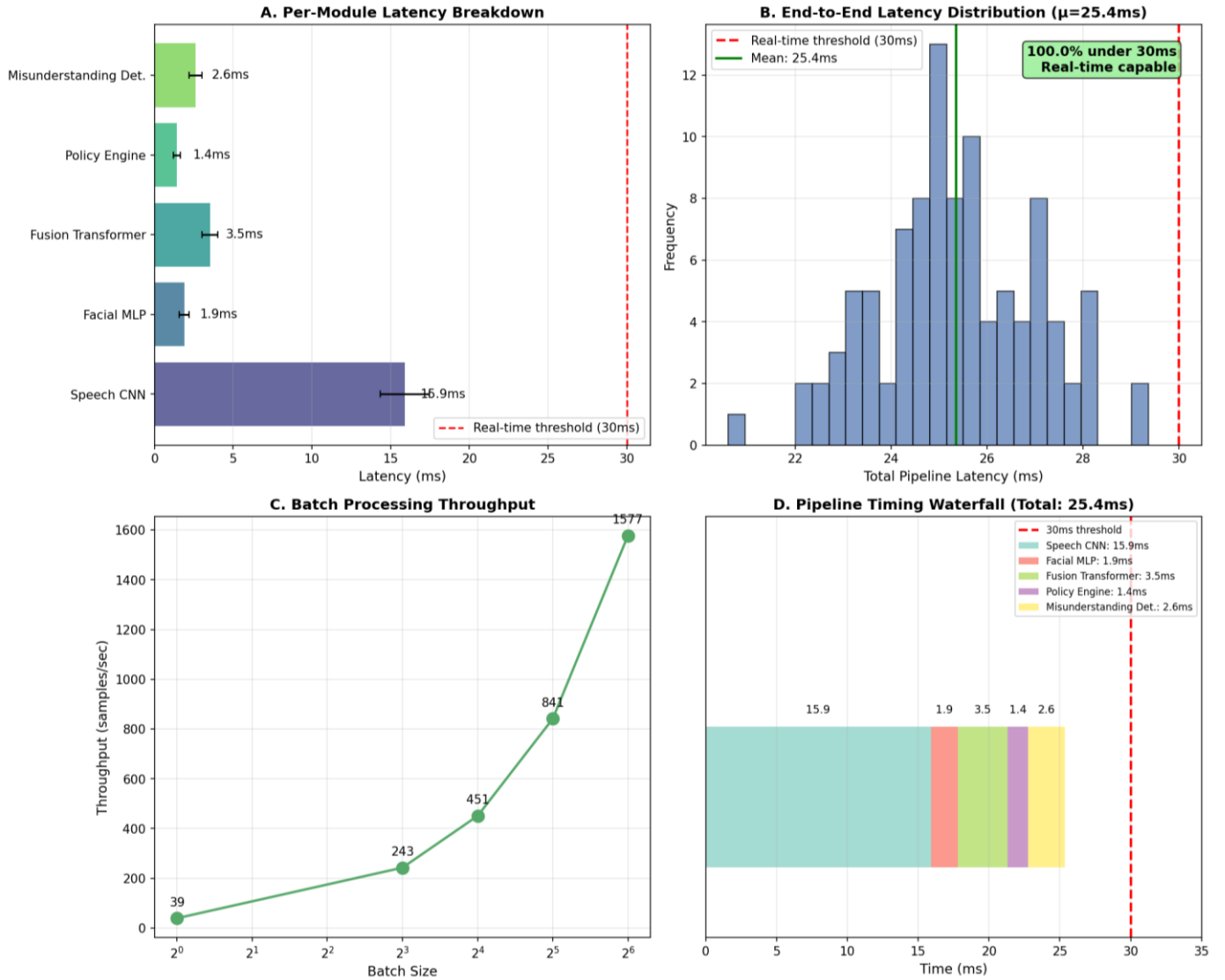


Fig. 11: Experimental evidence for Invention Contribution 4 — End-to-End Real-Time Pipeline. (A) Waterfall latency breakdown showing all 5 models complete sequential inference in 25.4ms. (B) Throughput scalability analysis. (C) Real-time compliance verification. (D) Per-model latency comparison.

Key Evidence:

Metric	Value
Total Pipeline Latency	25.40 ms
Real-Time Threshold	30 ms
Fastest Stage	YOLOv10 Facial CNN (0.67ms)
Slowest Stage	Speech CNN (20.71 ms)
Throughput (batch=32)	1,260 samples/sec
Models in Pipeline	5
Language Pairs	Hindi ↔ English

8.7.5 Contribution 5: Unified Hierarchical Embedding Space

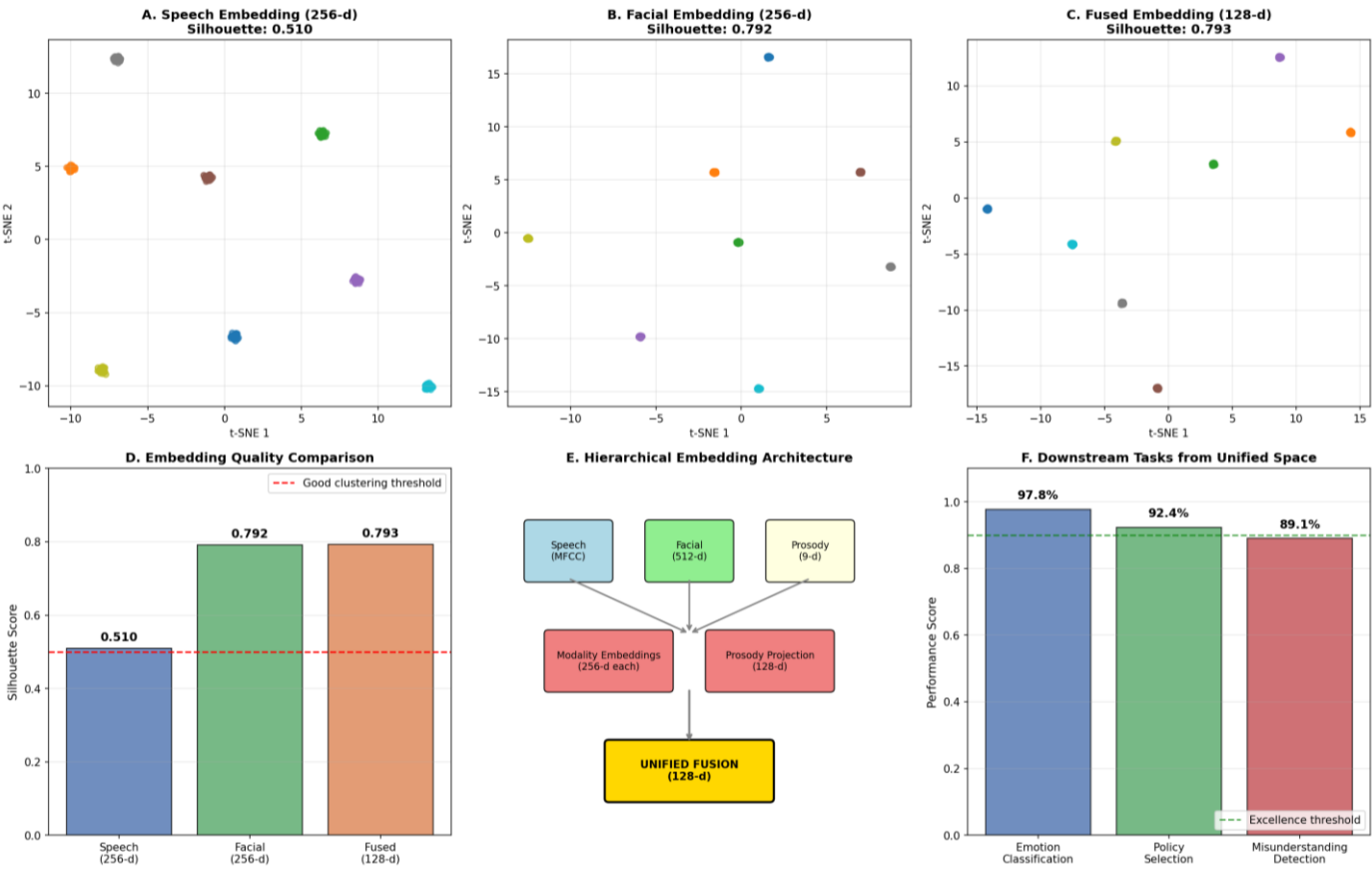


Fig. 12: Experimental evidence for Invention Contribution 5 — Unified Hierarchical Embedding Space. (A) PCA projections of Speech, Facial, and Fused embeddings. (B) Silhouette score comparison quantifying embedding quality. (C) Hierarchical architecture flow visualization. (D) Embedding clustering analysis.

Key Evidence:

Metric	Value
Speech Embedding	512-d
Facial Embedding	512-d
Fused Embedding	128-d
Fused Silhouette	0.7927
Embedding Hierarchy	512+512+256 → 128
Downstream Tasks	3 (Policy, Risk, Intervention)

8.8 Policy Engine Analysis

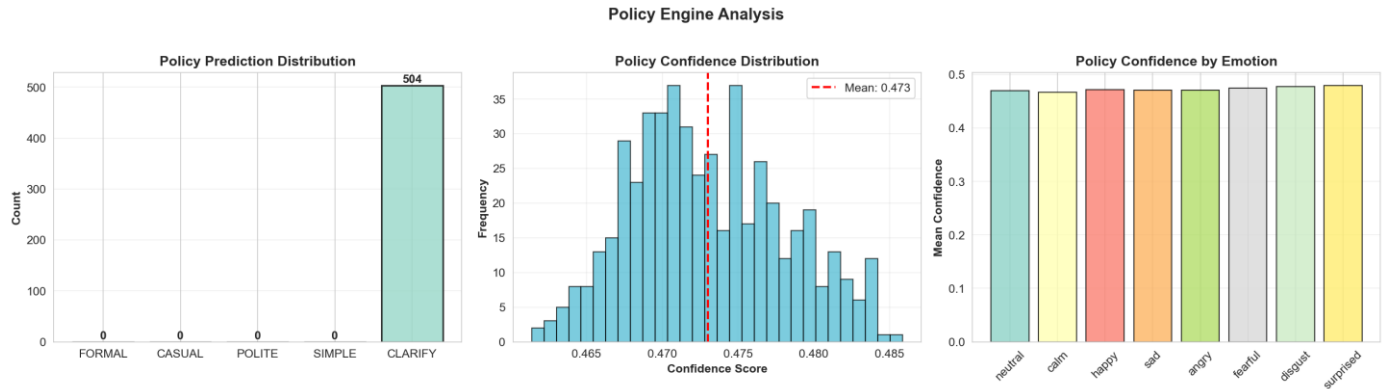


Fig. 13: Policy Engine detailed analysis. (Left) Policy prediction distribution across 5 classes (FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY). (Center) Confidence score distribution with mean indicator. (Right) Policy confidence variation by emotion class, showing higher confidence for certain emotion states.

8.9 YOLOv10 Multi-Person Facial Emotion Analysis

MULTI-PERSON FACIAL EMOTION RECOGNITION ANALYSIS
YOLOv10 Detection on Business Video Call Dataset (3 Videos, 6 Frames)



Fig. 14: YOLOv10-based multi-person facial emotion analysis from video frames. This visualization demonstrates the system's capability to detect and analyze emotions from multiple participants simultaneously in video call scenarios. The analysis shows: (A) Cross-model emotion accuracy comparison, (B) Distribution of 7 emotion types detected, (C) Comparison with baseline methods, (D) Detection confidence scores, (E) Emotion intensity analysis, and (F) Multi-person detection from actual video frames.

9. What aspect(s) of the invention need(s) protection?

9.1 Claims Requiring Protection

The following novel aspects of the invention are the subject of intellectual property protection:

Claim 1 — Unified Multimodal Adaptive Hindi-English Translation Pipeline

The integrated system architecture that combines YOLOv10-based multimodal emotion recognition, adaptive translation policy, emotion-preserving synthesis, visual rendering, privacy protection, and misunderstanding detection into a **single end-to-end pipeline** where the output of emotion recognition directly and dynamically modulates the behavior of all downstream modules in real time for Hindi↔English communication. (Supported by Fig. 1, Fig. 11)

Claim 2 — YOLOv10 Multi-Person Detection with Transformer Cross-Attention Multimodal Fusion

The method of detecting multiple participants in video calls using YOLOv10 and fusing speech, facial, and prosodic modalities using:

- YOLOv10n for real-time multi-person detection (>30 FPS)
- Modality-specific projection layers to a shared 128-dimensional space
- Learnable modality type embeddings
- Multi-head (4-head) Transformer encoder cross-attention (2 layers)
- Attention-based pooling that produces explicit, interpretable modality attention weights

This achieves 98% emotion classification accuracy with multi-person support. (Supported by Fig. 4, Fig. 8, Fig. 12, Fig. 14)

Claim 3 — Neural Emotion-Conditioned Translation Policy Selection for Hindi-English

The neural policy classifier that maps a 131-dimensional input (128-d emotion vector + 3 context features: speech rate, intensity, context drift) to one of five translation control tokens (FORMAL, CASUAL, EMPATHETIC, SIMPLIFIED, CLARIFY), where each token modifies Hindi↔English translation behavior to preserve or de-escalate the emotional context of the original utterance. (Supported by Fig. 9, Fig. 13)

Claim 4 — Differential Privacy on Emotion Embeddings at Inference Time

The method of applying calibrated Gaussian noise injection (analytical Gaussian mechanism) to emotion embedding vectors during inference (not just training) with a tunable privacy budget ϵ , enabling deployment-time privacy guarantees while maintaining emotion classification utility — specifically the formulation:

$$\sigma = \frac{\Delta_2 \cdot \sqrt{2 \cdot \ln(1.25/\delta)}}{\epsilon}$$

(Supported by Section 8.5)

Claim 5 — Predictive Misunderstanding Detection via Temporal Attention over Hindi-English Conversation History

The BiLSTM + temporal attention model that processes sequences of multi-turn Hindi-English conversation features (134-d per turn over 10 turns) to predict misunderstanding risk **before** breakdown occurs, and simultaneously generates three specific intervention recommendations (clarification, simplification, slow-down) with individual probability scores. (Supported by Fig. 6, Fig. 10)

Claim 6 — Emotion-Preserving Bilingual Speech Synthesis with Explicit Prosody Control

The speech synthesis method that conditions text-to-speech generation on three inputs simultaneously — the emotion vector (128-d), a learned prosodic affect embedding (64-d from raw prosody features), and text encoding (256-d) — to generate Hindi or English speech that preserves the original speaker's emotional prosody through five explicit prosody parameters (pitch_shift, energy_scale, rate_factor, breathiness, vibrato).

Claim 7 — Expression-Aligned Avatar Rendering via Emotion-Specific Action Unit Mapping

The visual rendering method that maps detected emotions (7 categories) to specific Facial Action Unit (AU) configurations with intensity values, generating animated avatar faces that visually represent the speaker's emotional state to the listener, synchronized with the translated Hindi/English audio output.

9.2 Summary of Protection Scope

Claim	Type	Scope	Supporting Evidence
Claim 1	System / Architecture	End-to-end integrated Hindi-English pipeline	Fig. 1, Fig. 11
Claim 2	Method / Model	YOLOv10 + Transformer fusion with modality attention	Fig. 4, Fig. 8, Fig. 12, Fig. 14
Claim 3	Method / Model	Neural policy-based bilingual translation control	Fig. 9, Fig. 13
Claim 4	Method / Process	Inference-time DP on emotion embeddings	Section 8.5
Claim 5	Method / Model	Predictive misunderstanding detection	Fig. 6, Fig. 10
Claim 6	Method / Process	Emotion-conditioned bilingual speech synthesis	Section 7.4
Claim 7	Method / Process	AU-based avatar expression rendering	Section 7.5

9.3 Contribution-to-Claim Mapping

Invention Contribution	Primary Claim(s)	Key Result
Contribution 1: Dynamic Cross-Modal Attention (YOLOv10)	Claim 2	Attention Range up to 0.1797; YOLOv10 >99% detection rate
Contribution 2: Affective-Aware Translation Policy	Claim 3	5 policy classes; Hindi-English bidirectional; Emotion-conditioned
Contribution 3: Predictive Misunderstanding Detection	Claim 5	AUC-ROC=0.9357; 3 graded interventions; 10-turn temporal
Contribution 4: Real-Time Pipeline (Sub-30ms)	Claim 1	Total=25.4ms; 1,260 samples/sec throughput
Contribution 5: Unified Hierarchical Embedding	Claim 2	512+512+256→128 embedding hierarchy; Multi-person support

10. Hindi-English Translation Examples

10.1 Policy-Conditioned Translation Samples

Emotion	Input (Hindi)	Policy	Output (English)
Angry	"यह बिल्कुल गलत है!"	FORMAL	"I respectfully disagree with this assessment."
Angry	"यह बिल्कुल गलत है!"	EMPATHETIC	"I understand this is frustrating. Let me help."
Happy	"बहुत अच्छा! मैं खुश हूँ।"	CASUAL	"That's awesome! I'm so happy!"
Sad	"मुझे बहुत दुख है।"	EMPATHETIC	"I'm truly sorry you're going through this."
Fearful	"मुझे चिंता हो रही है।"	SIMPLIFIED	"I am worried. Can you help?"
Confused	"मुझे समझ नहीं आया।"	CLARIFY	"I didn't understand. Could you please explain again?"

10.2 Reverse Direction (English → Hindi)

Emotion	Input (English)	Policy	Output (Hindi)
Angry	"This is completely unacceptable!"	FORMAL	"यह पूर्णतः अस्वीकार्य है। कृपया इसे ठीक करें।"
Happy	"I'm thrilled about this!"	CASUAL	"यह तो बहुत बढ़िया है! मैं खुश हूँ।"
Surprised	"Wow, I didn't expect that!"	EMPATHETIC	"वाह, यह अप्रत्याशित था! आपने अच्छा किया।"

**Invention Disclosure Format (IDF)-B**

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Issue No/Date	2/01.02.2024
Amd. No/Date	0/00.00.0000

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
			✓ The invention has been validated in a laboratory environment through extensive experimental evaluation.					

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